import pandas as pd import numpy as np import seaborn as sns import scipy.stats from scipy.stats import skew %matplotlib inline

import pandas as pd import numpy as np import seaborn as sns import scipy.stats from scipy.stats import skew %matplotlib inline

Task 1 - Importing all the required Libraries

```
import pandas as pd
import numpy as np
import seaborn as sns
import scipy.stats
from scipy.stats import skew
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use ('ggplot')
plt.rcParams['figure.figsize']= (12,8)
```

Task 2 - Load the data the advertisement dataset captures sales revenue generated with respect advertisement spend across multiple channels like TV, newspaper and radio

```
df = pd.read_csv('Advertising.csv')
df.head(10)
```

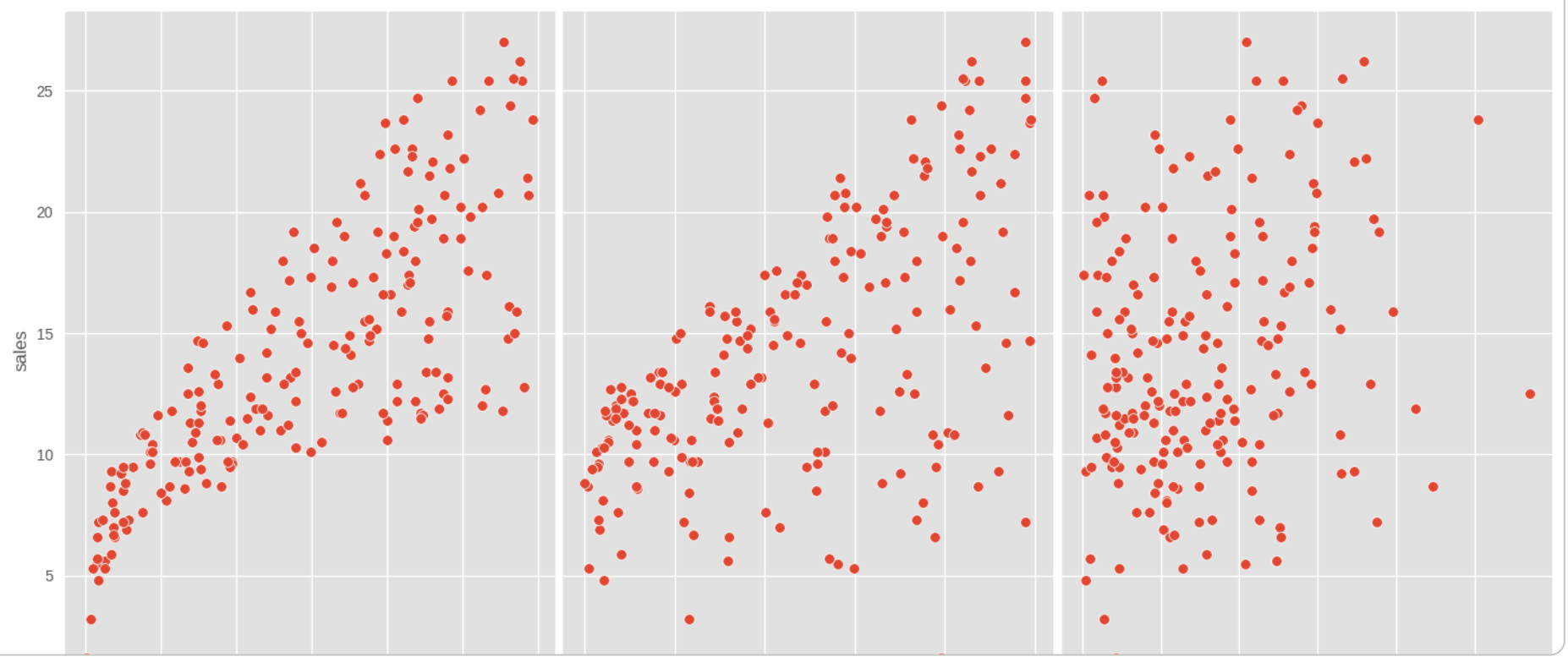
	TV	radio	newspaper	sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	9.3
3	151.5	41.3	58.5	18.5
4	180.8	10.8	58.4	12.9
5	8.7	48.9	75.0	7.2
6	57.5	32.8	23.5	11.8
7	120.2	19.6	11.6	13.2
8	8.6	2.1	1.0	4.8
9	199.8	2.6	21.2	10.6

Next steps: Generate code with df New interactive sheet

Task 3 - Relationship between feature and response

```
sns.pairplot(df, x_vars=['TV','radio','newspaper'], y_vars='sales', height=7, aspect=0.7 )
```

```
<seaborn.axisgrid.PairGrid at 0x7e2c133e6c30>
```



Task 4 - multiple linear regression model - Estimating Coefficient

```
from sklearn.linear_model import LinearRegression
x = df.drop('sales', axis=1)
y = df ['sales']
lm1 = LinearRegression()
lm1.fit(x,y)
print(lm1.intercept_)
print(lm1.coef_)
2.938889369459412
#store feature name
feature_name = ['TV', 'radio', 'newspaper']
#get regression coeffficients
coefficients = lm1.coef_
# creating dictionary to each column to its coefficients
feature_coefficient = dict(zip(feature_name, coefficients))
#displaying coefficient
feature_coefficient
{'TV': np.float64(0.0457646454553976),
'radio': np.float64(0.18853001691820448),
'newspaper': np.float64(-0.0010374930424763285)}
sns.heatmap(df.corr() , annot = True)
```

```
2.938889369459412
[ 0.04576465  0.18853002 -0.00103749]
<Axes: >
```

Task 5 - feature selection

```
from sklearn.metrics import r2_score
#model 1 using tv and radio
x_tv_radio = df [[ 'TV', 'radio']]
#initionailising model
lm2 = LinearRegression()
# train the model on full dataset
lm2.fit(x_tv_radio,y)
#predict the sales using trained model
lm2_pred = lm2.predict (x_tv_radio)
#Calculate R2 score for model evaluation
R2_tv_radio = r2_score(y,lm2_pred)
print("R2 score using TV and radio :" , R2_tv_radio)
#Model 2 using TV , radio newspaper
X_all_features = df[['TV' , 'radio' , 'newspaper']]
lm3 = LinearRegression()
lm3.fit(X_all_features,y)
lm3_pred = lm3.predict(X_all_features)
r2_all_features = r2_score(y,lm3_pred)
print("R2 score using TV, radio and newspaper;" , r2_all_features)
```

```
R2 score using TV and radio : 0.8971942610828957
R2 score using TV, radio and newspaper; 0.8972106381789522
```

Task 6 - model evaluation using train/test dataset here we are checking whether to use newsapeper as feature

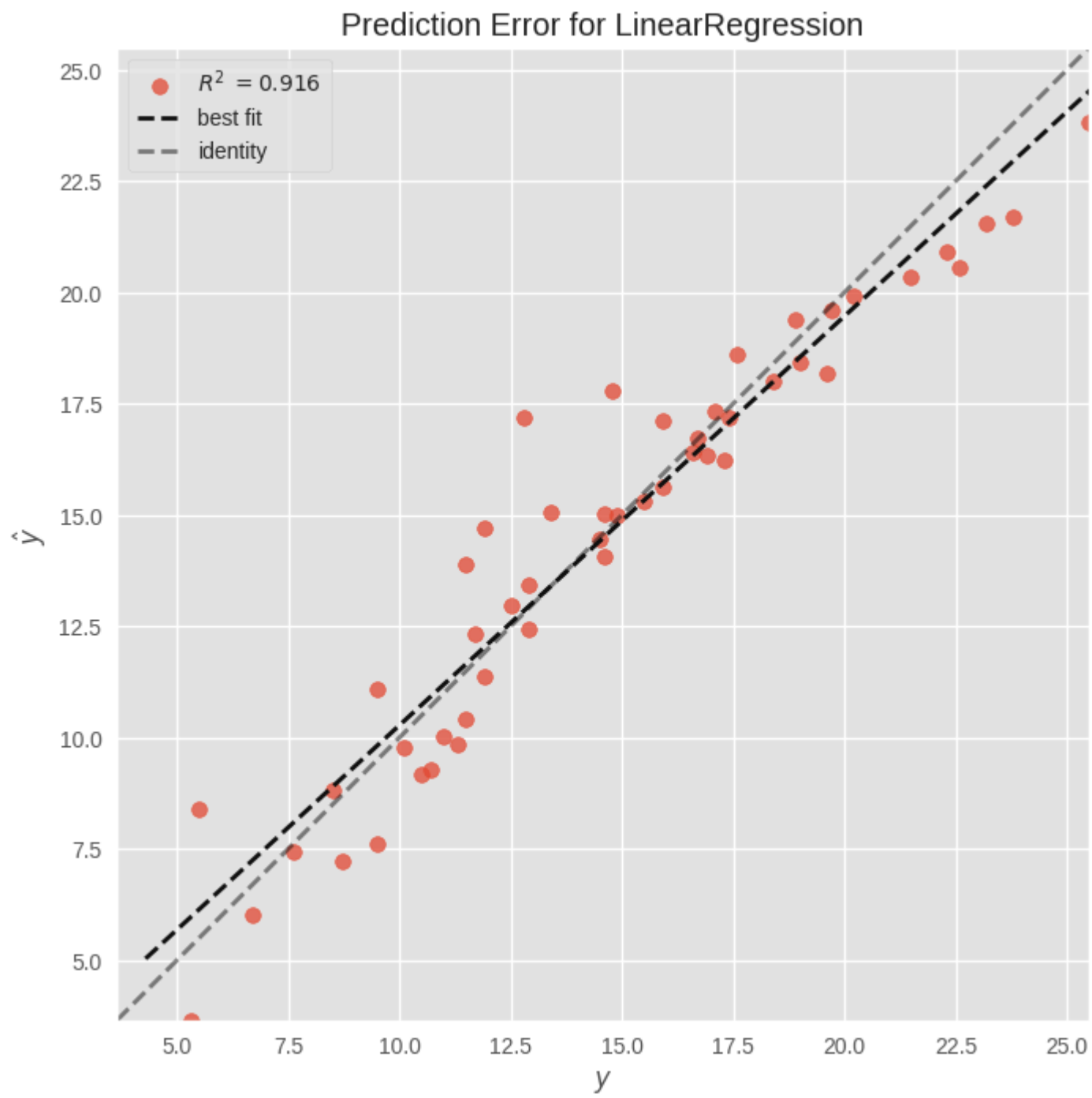
```
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error
X = df.drop('sales', axis=1)
Y = df['sales']
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=1)
lm4 = LinearRegression()
lm4.fit(X_train,Y_train)
lm4_pred = lm4.predict(X_test)
rmse = np.sqrt(mean_squared_error(Y_test,lm4_pred))
print(rmse)
r2_score = r2_score(Y_test,lm4_pred)
print(r2_score)
X = df.drop( ['sales' , 'newspaper'], axis=1)
y = df['sales']
from sklearn.metrics import r2_score
x_train, x_test, y_train, y_test = train_test_split(x,y, random_state=1)
lm5 = LinearRegression()
lm5.fit(x_train, y_train)
lm5_pred = lm5.predict(x_test)
rmse = np.sqrt(mean_squared_error(y_test, lm5_pred))
print(rmse)
r2_score = r2_score(y_test, lm5_pred)
print(r2_score)
```

```
1.4113417558581587
0.8927605914615384
1.404651423032895
0.9156213613792232
```

Task 7 - Model Diagnostics

```
from yellowbrick.regressor import ResidualsPlot, PredictionError
visualizer = PredictionError(lm5).fit(x_train, y_train)
visualizer.score(x_test, y_test)
visualizer.show()
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid
  warnings.warn(
```



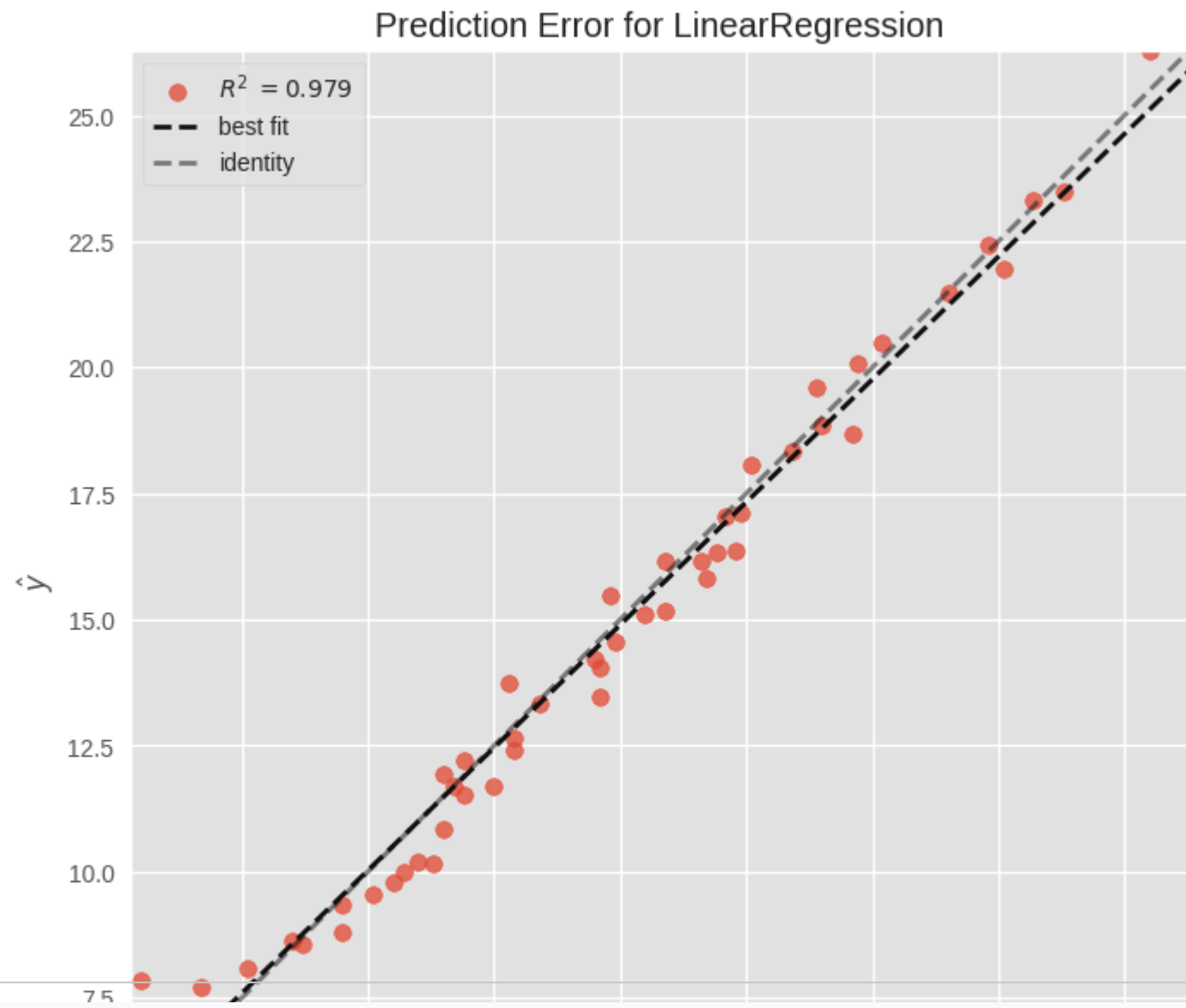

```
<Axes: title={'center': 'Prediction Error for LinearRegression'}, xlabel='$y$', ylabel='$\\hat{y}$'>
```

Task 8 - Interaction Effect (Synergy)

TV radio newspaper sales

```
#creating interation feature
from sklearn.metrics import r2_score
df['tv_radio_interaction'] = df['radio']*df['TV']
#select the features including interaction term
x = df[['TV','radio','tv_radio_interaction']]
y = df['sales']
x_train, x_test, y_train, y_test = train_test_split(x,y, random_state=1)
#Training the interaction model
interaction_model = LinearRegression()
interaction_model.fit(x_train, y_train)
y_test_predictions = interaction_model.predict(x_test)
print("RMSE :", np.sqrt ( mean_squared_error(y_test, y_test_predictions )))
print ( "R2 Score: " , r2_score(y_test, y_test_predictions))
#visualizing prediction performance
from sklearn.metrics import r2_score
from yellowbrick.regressor import ResidualsPlot, PredictionError
visualizer = PredictionError(interaction_model).fit(x_train, y_train)
visualizer.score(x_test, y_test)
visualizer.show()
```

```
RMSE : 0.7011871137164328
R2 Score:  0.978973681468126
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid
  warnings.warn(
```



Close

Conclusion

In this project, we analyzed the relationship between advertising spending and sales using the Advertising dataset. The dataset included three advertising channels: TV, Radio, and Newspaper. Through exploratory data analysis and visualization, we observed how different advertising platforms influence sales performance.

A Multiple Linear Regression model was developed to predict sales based on the advertising budget across these channels. The model coefficients showed that TV and Radio advertising have a stronger positive impact on sales, while Newspaper advertising has a relatively smaller effect.

To evaluate the model performance, metrics such as R-squared score and RMSE were used. The results indicate that the regression model can effectively predict sales with a good level of accuracy. The analysis also showed that combining TV and Radio advertising provides strong predictive performance.

Overall, this project demonstrates how machine learning techniques can be used to analyze marketing data and support

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Overall, this project demonstrates how machine learning techniques can be used to analyze marketing data and support data-driven decision making. Businesses can use such models to optimize their advertising budgets and improve sales performance.